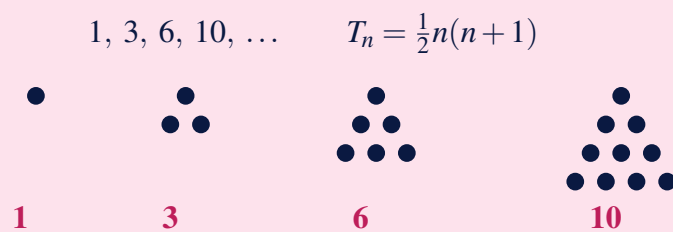


Sequences

Notes · Questions · Worked Answers

Linear n th term · Quadratic n th term · Cubic, fraction & geometric sequences ·
Subscript (term-to-term) formulae · Sums & special patterns · Diagram and
grid problems



How to use this booklet

- Read the pink **Key Notes** pages first.
- Section 1: linear sequences. Section 2: quadratic sequences. Section 3: cubic, fraction, geometric, sums and other styles.
- Each white **Question** page is followed straight away by a yellow **Answer** page — try every question before turning over.
- Work in the space given under each question.

Key Notes

1. Linear sequences (constant first difference)

n th term = $dn + c$ where d = common difference.

1. Find the difference d between terms.
2. Write dn (the “ d times table”).
3. Compare with the sequence: add or subtract to get c .
(c = the “zeroth term” = 1st term $- d$.)

Decreasing: d is negative, e.g.
19, 16, 13, ... $\Rightarrow 22 - 3n$.

Is x a term? Solve $dn + c = x$: whole-number $n \Rightarrow$ yes.

Example: 5, 9, 13, 17, ...

n	1	2	3	4
$4n$	4	8	12	16
term	5	9	13	17

each term is $4n + 1 \Rightarrow \boxed{4n + 1}$

Check: $n = 3$: $4(3) + 1 = 13 \checkmark$

2. Quadratic sequences (constant second difference)

n th term = $an^2 + bn + c$.

1. First differences, then second differences.
2. $a = \frac{\text{second difference}}{2}$.
3. Subtract an^2 from each term — what is left is **linear** ($bn + c$).
4. Find the linear n th term of what is left and add it on.

Short-cut: $a + b + c = 1$ st term, $3a + b = 1$ st difference.

Example: 2, 9, 20, 35, 54

1st diffs: 7, 11, 15, 19

2nd diffs: 4, 4, 4 $\Rightarrow a = 2$

term	2	9	20	35	54
$2n^2$	2	8	18	32	50
left	0	1	2	3	4

left = $n - 1 \Rightarrow \boxed{2n^2 + n - 1}$

3. Special sequences to recognise

Odd	1, 3, 5, 7, ...	$2n - 1$	Square	1, 4, 9, 16, ...	n^2
Even	2, 4, 6, 8, ...	$2n$	Cube	1, 8, 27, 64, ...	n^3
Triangular	1, 3, 6, 10, ...	$\frac{1}{2}n(n + 1)$	Powers of 2	1, 2, 4, 8, ...	2^{n-1}
Rectangular	2, 6, 12, 20, ...	$n(n + 1)$	Fibonacci-type	1, 1, 2, 3, 5, ...	add last two
Look for a link to these: e.g. 3, 10, 29, 66 is $n^3 + 2$; 0, 1, 4, 9 is $(n - 1)^2$; 2, 6, 12, 20 is $n^2 + n$.					

4. Geometric (exponential) sequences

Each term is **multiplied** by the same ratio r .

$$nth \text{ term} = a \times r^{n-1} \quad a = 1st \text{ term}, r = \frac{u_2}{u_1}$$

e.g. 15, 75, 375, ...: $a = 15, r = 5 \Rightarrow 15 \times 5^{n-1}$.

Fractional / negative ratio: $1, -\frac{1}{2}, \frac{1}{4}, \dots = \left(-\frac{1}{2}\right)^{n-1}$.

Solving $3^n = 6561$: keep multiplying by 3, or use $3^8 = 6561$ so $n = 8$.

Remember $2^0 = 1, 2^{-1} = \frac{1}{2}, 2^{-2} = \frac{1}{4}$.

5. Cubic sequences

Third differences are constant. In IGCSE they are almost always linked to n^3 : subtract n^3 (1, 8, 27, 64, ...) and look at what is left. e.g. 2, 10, 30, 68: $-n^3$ leaves 1, 2, 3, 4 $\Rightarrow n^3 + n$.

6. Fraction sequences

Treat the **numerators** and **denominators** as two separate sequences, find the n th term of each, then write $\frac{\text{numerator } nth \text{ term}}{\text{denominator } nth \text{ term}}$. e.g. $\frac{1}{4}, \frac{2}{5}, \frac{3}{6}, \dots = \frac{n}{n+3}$. A denominator that doubles is $k \times 2^n$.

7. Combined sequences (“use your answers to part (a)”)

A new sequence is often the sum, difference or product of earlier ones. Line the terms up and test: $5, 16, 39, 80 = (1, 8, 27, 64) + (4, 8, 12, 16) \Rightarrow n^3 + 4n$; $-3, 0, 5, 12 = (n+2)(n-2) = n^2 - 4$.

8. Subscript (term-to-term) formulae

u_n means the n th term; u_{n-1} is the term before, u_{n-2} the one before that.

e.g. $u_1 = 3$, $u_2 = 4$, $u_n = 2u_{n-1} + u_{n-2}$: $u_3 = 2(4) + 3 = 11$, $u_4 = 2(11) + 4 = 26, \dots$

To go **backwards**, rearrange: if $u_n = 2u_{n-2} + u_{n-1}$ then $u_{n-2} = \frac{1}{2}(u_n - u_{n-1})$.

9. Sums of sequences

$$1 + 2 + \dots + n = \frac{1}{2}n(n+1)$$

$$2 + 4 + \dots + 2n = n(n+1)$$

$$1 + 3 + \dots + (2n-1) = n^2$$

$$1^3 + 2^3 + \dots + n^3 = \left[\frac{1}{2}n(n+1)\right]^2$$

Sum from 121 to 200 = $S_{200} - S_{120}$.

To show $S_n - S_{n-1} = P_n$: write both, take out common factors, simplify.

10. Diagram problems & “find p and q ”

Count carefully from the diagrams and fill the table first — then find the n th term of each row.

When a formula with unknowns is given (e.g. $\frac{1}{3}n^3 + pn^2 + qn$): **substitute** $n = 1$ and $n = 2$ using values from the table, then solve the two **simultaneous equations**. You do *not* need the difference method.

11. Grid (algebraic proof) questions

In a grid with t columns, the number below n is $n + t$ and the number to the right is $n + 1$. Write every cell in terms of n , expand the brackets and simplify — the n^2 and n terms cancel to leave a constant.

- 1 Determine the *formula* for each of the following sequences:
- (a) 6, 10, 14, 18, 22, ...
 - (b) 11, 13, 15, 17, 19, ...
 - (c) 9, 16, 23, 30, 37, ...
 - (d) 34, 56, 78, 100, 122, ...
 - (e) 22, 31, 40, 49, 58, ...
- 2 Determine the *formula* that generates each of the following sequences:
- (a) 0, -5, -10, -15, -20, ...
 - (b) -18, -16, -14, -12, -10, ...
 - (c) -5, -8, -11, -14, -17, ...
 - (d) 8, 1, -6, -13, -20, ...
 - (e) -7, -3, 1, 5, 9, ...

1 (a) $4n + 2$ (b) $2n + 9$ (c) $7n + 2$ (d) $22n + 12$ (e) $9n + 13$

Method: difference d gives dn ; then adjust using the 1st term, e.g. (a) $4n$: 4, 8, ... need +2.

2 (a) $5 - 5n$ (b) $2n - 20$ (c) $-3n - 2$ (d) $15 - 7n$ (e) $4n - 11$

- 3 Determine the *general formula* for each of the following sequences:
- (a) 1, 4, 7, 10, 13, ... (b) 2, 6, 10, 14, 18, ...
(c) 4, 13, 22, 31, 40, ... (d) 5, 15, 25, 35, 45, ...
(e) 1, 20, 39, 58, 77, ...
5. For the sequence,
18, 16, 14, 12, 10, ...
- (a) calculate the *difference* between successive terms,
(b) determine the *formula* that generates the sequence.
6. Determine the *general formula* for each of the following sequences:
- (a) 19, 16, 13, 10, 7, ... (b) 100, 96, 92, 88, 84, ...
(c) 41, 34, 27, 20, 13, ... (d) 66, 50, 34, 18, 2, ...
(e) 90, 81, 72, 63, 54, ...

3 Q4 (a) $3n - 2$ (b) $4n - 2$ (c) $9n - 5$ (d) $10n - 5$ (e) $19n - 18$

Q5 (a) difference = -2 (b) $20 - 2n$

Q6 (a) $22 - 3n$ (b) $104 - 4n$ (c) $48 - 7n$ (d) $82 - 16n$ (e) $99 - 9n$

4The n th term of a sequence is $8n - 3$.

- (i) Write down the first two terms of this sequence.

..... [1]

- (ii) Show that the number 203 is not in this sequence.

[2]

- (b) Find the
- n
- th term of these sequences.

- (i) 13, 19, 25, 31, ...

..... [2]

- (ii) 4, 8, 14, 22, ...

..... [2]

- (c) ... , 20, 50, ...

The second term of this sequence is 20 and the third term is 50.

The rule for finding the next term in this sequence is subtract y then multiply by 5.Find the value of y and work out the first term of this sequence. $y =$

First term = [4]

4 (a)(i) $8(1) - 3 = 5$, $8(2) - 3 = \boxed{5, 13}$ (ii) $8n - 3 = 203 \Rightarrow n = 25.75$, not a whole number, so 203 is not a term.

(b)(i) $\boxed{6n + 7}$ (ii) differences 4, 6, 8; 2nd diff 2 $\Rightarrow n^2$; $4 - 1 = 3$, $8 - 4 = 4$, $14 - 9 = 5 \Rightarrow n + 2$: $\boxed{n^2 + n + 2}$

(c) $(20 - y) \times 5 = 50 \Rightarrow \boxed{y = 10}$; $(t - 10) \times 5 = 20 \Rightarrow$ first term $\boxed{14}$

5

(a) Write down the 10th term and the n th term of the following sequences.

(i) 1, 2, 3, 4, 5 ..., ..., [1]

(ii) 7, 8, 9, 10, 11 ..., ..., [1]

(iii) 8, 10, 12, 14, 16 ..., [3]

(b) Consider the sequence

1(8 - 7), 2(10 - 8), 3(12 - 9), 4(14 - 10),,

(i) Write down the next term and the 10th term of this sequence in the form $a(b - c)$ where a , b and c are integers. [3]

(ii) Write down the n th term in the form $a(b - c)$ and then simplify your answer. [2]

5 (a)(i) 10th = $\boxed{10}$, $\boxed{n}$ (ii) $\boxed{16}$, $\boxed{n+6}$ (iii) $\boxed{26}$, $\boxed{2n+6}$
(b)(i) next = $\boxed{5(16-11)}$; 10th = $\boxed{10(26-16)}$ (ii) $n((2n+6)-(n+6)) = n \times n =$
 $\boxed{n^2}$

- 6** Determine the formula for the general term of each of the following sequences:
- (a) 1, 7, 17, 31, 49, 71, ...
 - (b) 6, 18, 38, 66, 102, 146, ...
 - (c) -5, 10, 35, 70, 115, 170, ...
 - (d) 1, 10, 25, 46, 73, 106, ...
-
- 7** A sequence is listed below:
2, 9, 20, 35, 54, 77, ...
- (a) Calculate the second differences for this sequence.
 - (b) Form a simpler sequence by subtracting $2n^2$ from each term.
 - (c) Determine a formula for the general term of the simpler sequence.
 - (d) Determine a formula for the general term of the original sequence.

6 (a) $2n^2 - 1$ (b) $4n^2 + 2$ (c) $5n^2 - 10$ (d) $3n^2 - 2$

e.g. (b): differences 12, 20, 28, 2nd difference 8 $\Rightarrow 4n^2$; $6 - 4 = 2$, $18 - 16 = 2 \Rightarrow +2$.

7 (a) 1st diffs 7, 11, 15, 19, 23; 2nd diffs $4, 4, 4, 4$ (b) $2 - 2, 9 - 8, 20 - 18, \dots =$
 $0, 1, 2, 3, 4$

(c) $n - 1$ (d) $2n^2 + n - 1$

- 8** (a) Calculate the second differences of the sequence,
6, 17, 36, 63, 98, 141, ...
- (b) Determine the formula for the general term of the sequence.

- 9** Determine the formula for the general term of each of the following sequences:

(a) 3, 17, 39, 69, 107, ...

(b) 5, 18, 37, 62, 93, ...

(c) 9, 23, 45, 75, 113, ...

(d) -4 , 12, 38, 74, 120, ...

8 (a) 1st diffs 11, 19, 27, 35, 43; 2nd diff $\boxed{8}$ (b) $6 - 4, 17 - 16, 36 - 36 = 2, 1, 0 \Rightarrow$
 $3 - n$: $\boxed{4n^2 - n + 3}$

9 (a) $\boxed{4n^2 + 2n - 3}$ (b) $\boxed{3n^2 + 4n - 2}$ (c) $\boxed{4n^2 + 2n + 3}$ (d) $\boxed{5n^2 + n - 10}$

10

Complete the table for the four sequences A, B, C and D.

	Sequence				Next term	n th term
A	2	5	8	11		
B	20	14	8	2		
C	1	4	9	16		
D	0	2	6	12		

[10]

(b) The sum of the first n terms of a sequence is $\frac{n(3n+1)}{2}$.

(i) When the sum of the first n terms is 155, show that $3n^2 + n - 310 = 0$.

[2]

(ii) Solve $3n^2 + n - 310 = 0$.

$n = \dots\dots\dots$ or $n = \dots\dots\dots$ [3]

(iii) Complete the statement.

The sum of the first $\dots\dots\dots$ terms of this sequence is 155.

[1]

10 (a) A: $\boxed{14}$, $\boxed{3n-1}$; B: $\boxed{-4}$, $\boxed{26-6n}$; C: $\boxed{25}$, $\boxed{n^2}$; D: $\boxed{20}$, $\boxed{n^2-n}$

(b)(i) $\frac{n(3n+1)}{2} = 155 \Rightarrow 3n^2 + n = 310 \Rightarrow 3n^2 + n - 310 = 0 \checkmark$

(ii) $(3n+31)(n-10) = 0 \Rightarrow n = \boxed{10}$ or $\boxed{-\frac{31}{3}}$ (iii) $\boxed{10}$

- 11** (b) (i) Write the four missing terms in the table for sequences A, B, C and D.

Term	1	2	3	4	5		n
Sequence A	-4		2	5	8		$3n - 7$
Sequence B	1	4	9	16	25		
Sequence C	5	10	15	20	25		
Sequence D	6	14	24	36	50		

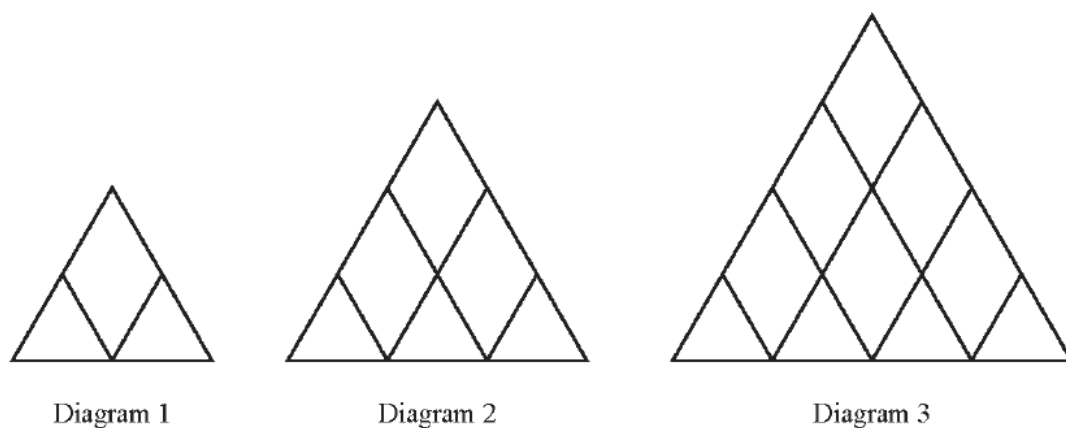
- (ii) Which term in sequence D is equal to 500?

[4]

Answer(b)(ii) [2]

- 11** (i) A: $\boxed{-1}$; B: $\boxed{n^2}$; C: $\boxed{5n}$; D = B + C: $\boxed{n^2 + 5n}$
(ii) $n^2 + 5n = 500 \Rightarrow (n + 25)(n - 20) = 0 \Rightarrow \boxed{20\text{th term}}$

12



Each diagram is made from tiles in the shape of equilateral triangles and rhombuses.
The length of a side of each tile is 1 unit.

(a) Complete the table below for this sequence of diagrams.

Diagram	1	2	3	4	5
Number of equilateral triangle shaped tiles	2	3	4	5	6
Number of rhombus shaped tiles	1	3	6		
Total number of tiles	3	6	10		
Number of 1 unit lengths	8	15	24		

[6]

(b) (i) The number of 1 unit lengths in Diagram n is $n^2 + 4n + p$.

Find the value of p .

$p = \dots\dots\dots$ [2]

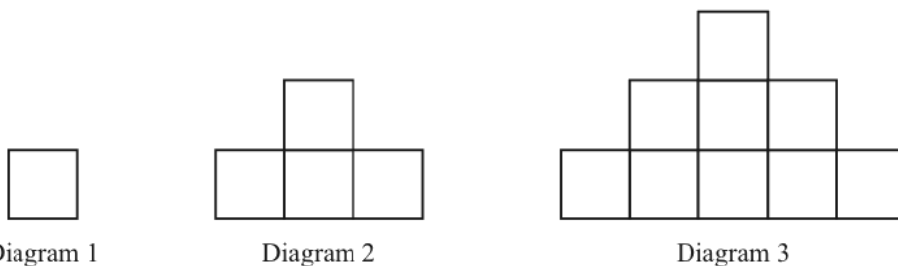
(ii) Calculate the number of 1 unit lengths in Diagram 10.

$\dots\dots\dots$ [1]

12 (a) Diagram 4: 10 rhombus, 15 total, 35 lengths; Diagram 5: 15, 21, 48

(b)(i) $n = 1$: $1 + 4 + p = 8 \Rightarrow \boxed{p = 3}$ (ii) $100 + 40 + 3 = \boxed{143}$

- 13** The first three diagrams in a sequence are shown below.
The diagrams are made by drawing lines of length 1 cm.



- (a) The areas of each of the first three diagrams are shown in this table.

Diagram	1	2	3
Area (cm ²)	1	4	9

- (i) Find the area of Diagram 4.

Answer(a)(i) cm² [1]

- (ii) Find, in terms of n , the area of Diagram n .

Answer(a)(ii) cm² [1]

- (b) The numbers of 1 cm lines needed to draw each of the first three diagrams are shown in this table.

Diagram	1	2	3
Number of 1 cm lines	4	13	26

- (i) Find the number of 1 cm lines needed to draw Diagram 4.

Answer(b)(i) [1]

- (ii) In which diagram are 118 lines of length 1 cm needed?

Answer(b)(ii) [1]

- (c) The **total** number of 1 cm lines needed to draw both Diagram 1 and Diagram 2 is 17.
The **total** number of 1 cm lines needed to draw all of the first n diagrams is

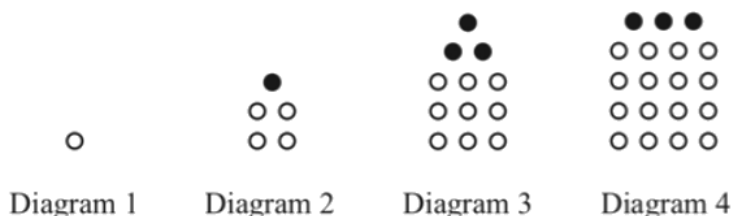
$$\frac{2}{3}n^3 + an^2 + bn.$$

Question continues on the next page →

Find the value of a and the value of b .
Show all your working.

- 13** (a)(i) $\boxed{16}$ (ii) $\boxed{n^2}$ (b)(i) differences 9, 13, (17): $\boxed{43}$ (ii) $2n^2 + 3n - 1 = 118 \Rightarrow (2n + 17)(n - 7) = 0$: $\boxed{7}$
(c) $n = 1$: $\frac{2}{3} + a + b = 4$; $n = 2$: $\frac{16}{3} + 4a + 2b = 17$. So $a + b = \frac{10}{3}$, $2a + b = \frac{35}{6}$
 $\Rightarrow \boxed{a = \frac{5}{2}, b = \frac{5}{6}}$

14



These are the first four diagrams of a sequence.
The diagrams are made from white dots and black dots.

(a) Complete the table for Diagram 5 and Diagram 6.

Diagram	1	2	3	4	5	6
Number of white dots	1	4	9	16		
Number of black dots	0	1	3	6		
Total number of dots	1	5	12	22		

[2]

(b) Write an expression, in terms of n , for the number of white dots in Diagram n .

..... [1]

(c) The expression for the total number of dots in Diagram n is $\frac{1}{2}(3n^2 - n)$.

(i) Find the total number of dots in Diagram 8.

..... [1]

(ii) Find an expression for the number of black dots in Diagram n .
Give your answer in its simplest form.

..... [2]

14 (a) D5: 25, 10, 35; D6: 36, 15, 51 (b) n^2

(c)(i) $\frac{1}{2}(3 \times 64 - 8) = 92$ (ii) $\frac{1}{2}(3n^2 - n) - n^2 = \frac{1}{2}(n^2 - n)$

15

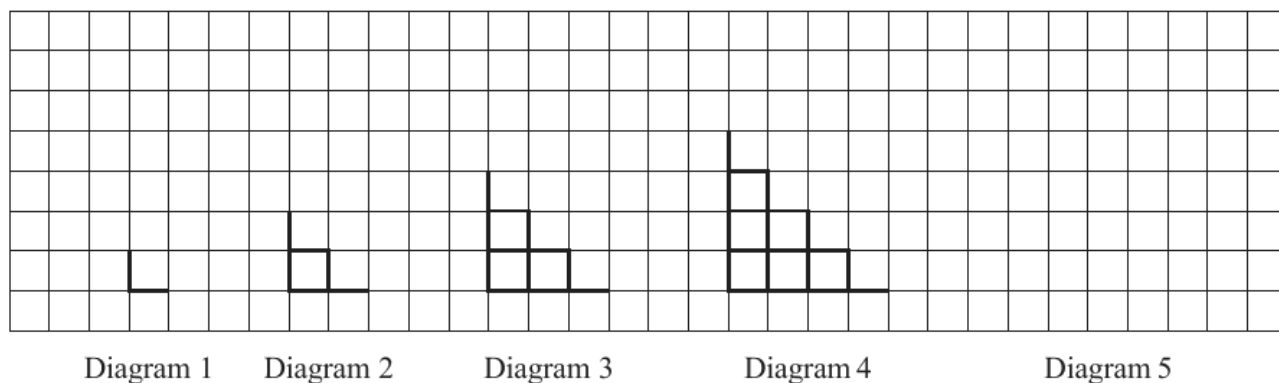


Diagram 1 shows two lines of length 1 unit at right angles forming an L .

Two L s are added to Diagram 1 to make Diagram 2. This forms one small square.

Three L s are added to Diagram 2 to make Diagram 3. This forms three small squares.
The sequence of Diagrams continues.

(a) Draw Diagram 5.

[1]

(b) Complete the table.

	Diagram 1	Diagram 2	Diagram 3	Diagram 4	Diagram 5
Number of lines of length 1 unit	2	6	12	20	
Number of small squares	0	1	3	6	

[2]

(c) Find an expression, in terms of n , for the number of lines of length 1 unit in Diagram n .

[2]

(d) Find an expression, in terms of n , for the number of small squares in Diagram n .

[2]

15 (a) Diagram 5: add five L-shapes to Diagram 4 (a staircase 5 squares wide at the base).

(b) Diagram 5: $\boxed{30}$ lines, $\boxed{10}$ squares (c) $\boxed{n(n+1)}$ (d) $\boxed{\frac{1}{2}n(n-1)}$

16

For
Examiner's
Use

Diagram 1

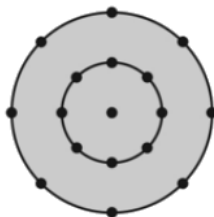


Diagram 2

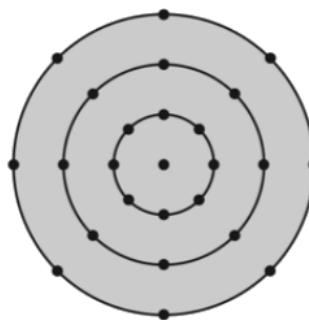


Diagram 3

The diagrams show a sequence of dots and circles.
 Each diagram has one dot at the centre and 8 dots on each circle.
 The radius of the first circle is 1 unit.
 The radius of each new circle is 1 unit greater than the radius of the previous circle.

(a) Complete the table for diagrams 4 and 5.

Diagram	1	2	3	4	5
Number of dots	9	17	25		
Area of the largest circle	π	4π	9π		
Total length of the circumferences of the circles	2π	6π	12π		

[4]

(b) (i) Write down, in terms of n , the number of dots in diagram n .

Answer(b)(i) [2]

(ii) Find n , when the number of dots in diagram n is 1097.

Answer(b)(ii) $n =$ [2]

(c) Write down, in terms of n and π , the area of the largest circle in

(i) diagram n ,

Answer(c)(i) [1]

(ii) diagram $3n$.

Answer(c)(ii) [1]

(d) Find, in terms of n and π , the total length of the circumferences of the circles in diagram n .

Answer(d) [2]

- 16** (a) D4: $33, 16\pi, 20\pi$; D5: $41, 25\pi, 30\pi$ (b)(i) $8n + 1$ (ii) $8n + 1 = 1097$:
 137
- (c)(i) πn^2 (ii) $9\pi n^2$ (d) $2\pi(1 + 2 + \cdots + n) = \pi n(n + 1)$

17

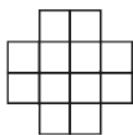


Diagram 1

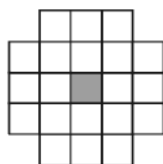


Diagram 2

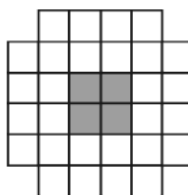


Diagram 3

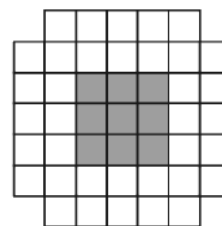


Diagram 4

The first four diagrams in a sequence are shown above.

The diagrams are drawn using white squares  and grey squares .

(a) Complete the columns in the table for Diagram 4 and Diagram n .

Diagram	1	2	3	4		n
Number of white squares	12	20	28			
Number of grey squares	0	1	4			
Total number of squares	12	21	32			$(n+1)(n+5)$

[6]

(b) Work out the number of the diagram which has a total of 480 squares.

Answer(b) [2]

(c) The **total** number of squares in the **first n diagrams** is

$$\frac{1}{3}n^3 + pn^2 + qn.$$

(i) Use $n = 1$ in this expression to show that $p + q = 11\frac{2}{3}$.

Answer(c)(i)

[1]

(ii) Use $n = 2$ in the expression to show that $4p + 2q = 30\frac{1}{3}$.

Question continues on the next page →

Answer(c)(ii)

[2]

- (iii) Find the values of p and q .

Answer(c)(iii) $p = \dots\dots\dots$ $q = \dots\dots\dots$ [3]

- 17** (a) Diagram 4: 36, 9, 45; Diagram n : white $8n + 4$, grey $(n - 1)^2$
 (b) $(n + 1)(n + 5) = 480 \Rightarrow n^2 + 6n - 475 = 0 \Rightarrow (n - 19)(n + 25) = 0$: 19
 (c)(i) $\frac{1}{3} + p + q = 12 \Rightarrow p + q = 11\frac{2}{3} \checkmark$ (ii) $\frac{8}{3} + 4p + 2q = 12 + 21 = 33 \Rightarrow 4p + 2q = 30\frac{1}{3} \checkmark$
 (iii) $2p + 2q = 23\frac{1}{3}$; subtract: $2p = 7 \Rightarrow p = 3.5$, $q = \frac{49}{6}$

18

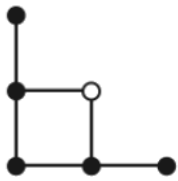
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Diagram 1

1 white dot

5 black dots

6 lines

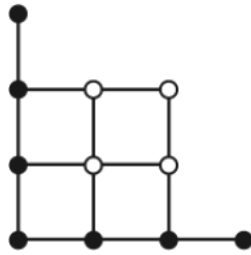


Diagram 2

4 white dots

7 black dots

14 lines

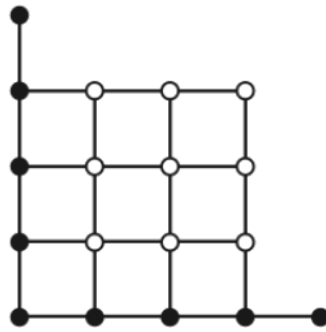


Diagram 3

9 white dots

9 black dots

26 lines

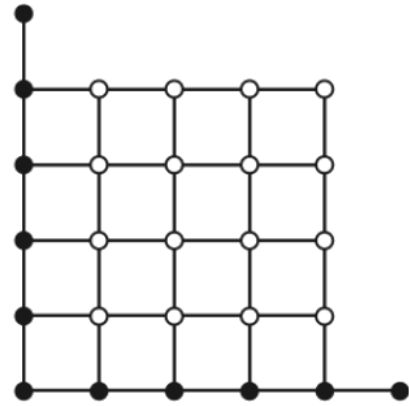


Diagram 4

16 white dots

11 black dots

42 lines

The four diagrams above are the first four of a pattern.

(a) Diagram 5 has been started below.

Complete this diagram and write down the information about the numbers of dots and lines.

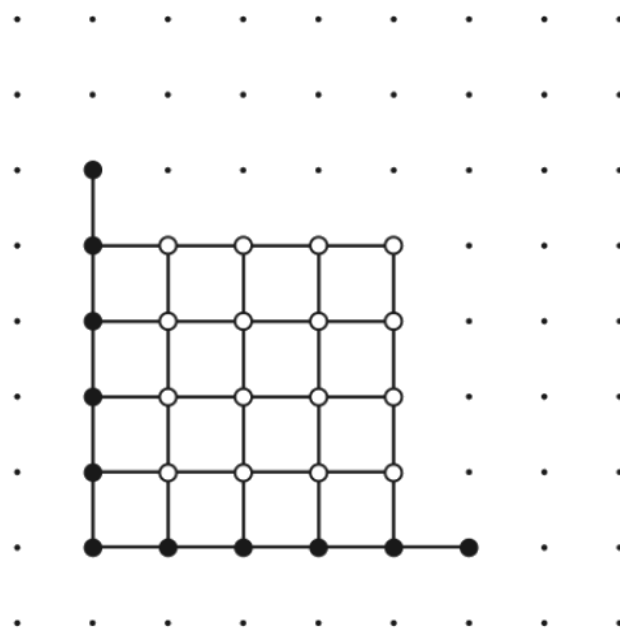


Diagram 5

..... white dots

..... black dots

..... lines

[4]

Question continues on the next page →

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- (b) Complete the information about the number of dots and lines in Diagram 8.

Answer(b) white dots
..... black dots
..... lines [3]

- (c) Complete the information about the number of dots in Diagram n .
Give your answers in terms of n .

Answer(c) white dots
..... black dots [2]

- (d) The number of lines in diagram n is $k(n^2 + n + 1)$.

Find

- (i) the value of k ,

Answer(d)(i) $k =$ [1]

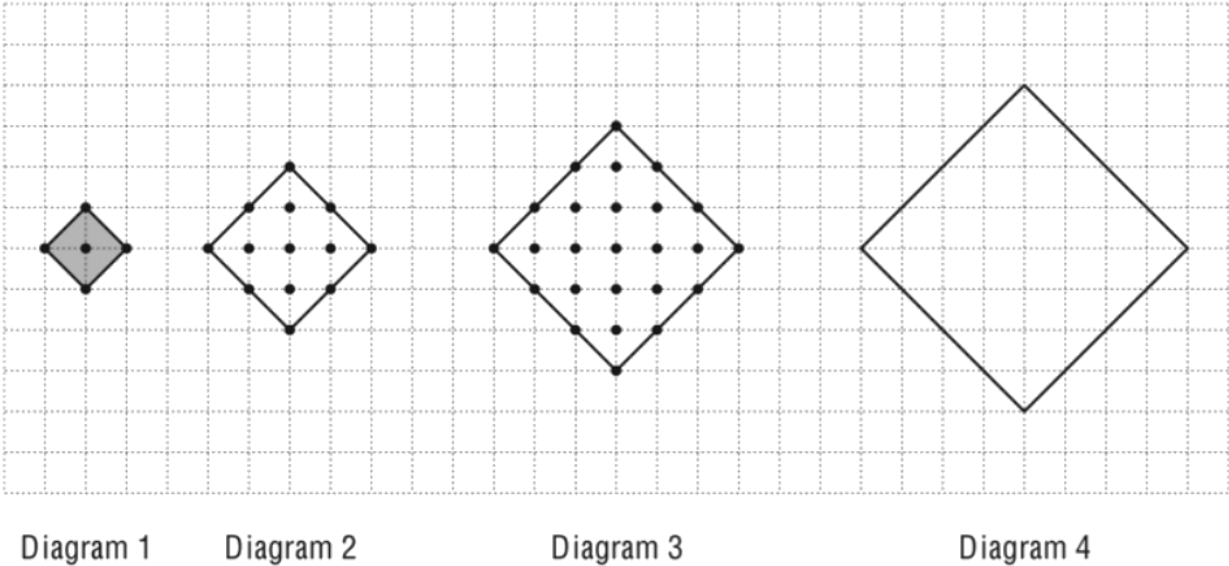
- (ii) the number of lines in Diagram 100.

Answer(d)(ii) [1]

- 18** (a) Diagram 5: $\boxed{25}$ white, $\boxed{13}$ black, $\boxed{62}$ lines (b) Diagram 8: $\boxed{64}$ white, $\boxed{19}$ black, $\boxed{146}$ lines
- (c) white $\boxed{n^2}$, black $\boxed{2n+3}$ (d)(i) $k(1+1+1) = 6 \Rightarrow \boxed{k=2}$ (ii) $2(10000+100+1) = \boxed{20202}$

19

Ex



The diagrams show squares and dots on a grid.

Some dots are on the sides of each square and other dots are inside each square.

The area of the square (shaded) in Diagram 1 is 1 unit².

- (a) Complete Diagram 4 by marking all the dots. [1]
- (b) Complete the columns in the table below for Diagrams 4, 5 and n .

Diagram	1	2	3	4	5	-----	n
Number of units of area	1	4	9			-----	
Number of dots inside the square	1	5	13			-----	$(n - 1)^2 + n^2$
Number of dots on the sides of the square	4	8	12			-----	
Total number of dots	5	13	25			-----	

- (c) For Diagram 200, find the number of dots
- (i) inside the square,

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Use

Answer(c)(i) [1]

(ii) on the sides of the square.

Answer(c)(ii) [1]

(d) Which diagram has 265 dots inside the square?

Answer(d) [1]

19 (a) Diagram 4: 16 dots on the sides, 25 inside (a 4×4 diamond pattern).

(b) D4: 16, 25, 16, 41; D5: 25, 41, 20, 61; n : n^2 , $(n-1)^2 + n^2$, $4n$, $n^2 + (n+1)^2$

(c)(i) $199^2 + 200^2 = 79601$ (ii) 800 (d) $(n-1)^2 + n^2 = 265 \Rightarrow n^2 - n - 132 = 0$:

12

20 --

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Examiner's
Use


Diagram 1

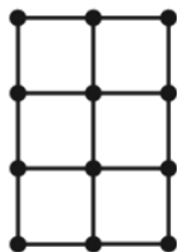


Diagram 2

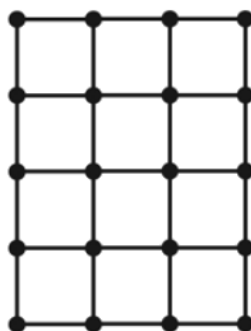


Diagram 3

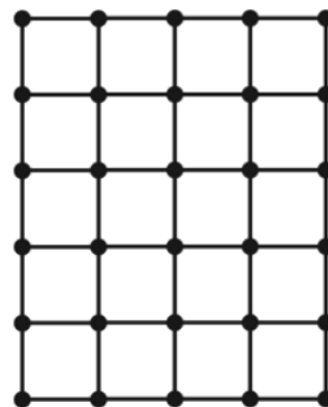


Diagram 4

The first four Diagrams in a sequence are shown above.
Each Diagram is made from dots and one centimetre lines.
The area of each small square is 1 cm^2 .

(a) Complete the table for Diagrams 5 and 6.

Diagram	1	2	3	4	5	6
Area (cm^2)	2	6	12	20		
Number of dots	6	12	20	30		
Number of one centimetre lines	7	17	31	49		

[4]

(b) The **area** of Diagram n is $n(n+1) \text{ cm}^2$.

(i) Find the **area** of Diagram 50.

Answer(b)(i) cm^2 [1]

(ii) Which Diagram has an **area** of 930 cm^2 ?

Answer(b)(ii) [1]

(c) Find, in terms of n , the number of **dots** in Diagram n .

Answer(c) [1]

Question continues on the next page →

(d) The number of one centimetre lines in Diagram n is $2n^2 + pn + 1$.

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Use

(i) Show that $p = 4$.

Answer(d)(i)

[2]

(ii) Find the number of one centimetre lines in Diagram 10.

Answer(d)(ii) [1]

(iii) Which Diagram has 337 one centimetre lines?

Answer(d)(iii) [3]

(e) For **each** Diagram, the number of squares of area 1 cm^2 is A , the number of dots is D and the number of one centimetre lines is L .

Find a connection between A , D and L that is true for each Diagram.

Answer(e) [1]

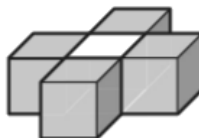
- 20** (a) D5: 30, 42, 71; D6: 42, 56, 97 (b)(i) $50 \times 51 = \boxed{2550}$ (ii) $n(n+1) = 930$:
 $\boxed{30}$
- (c) $\boxed{(n+1)(n+2)}$ (d)(i) $n = 1: 2 + p + 1 = 7 \Rightarrow p = 4 \checkmark$ (ii) $200 + 40 + 1 = \boxed{241}$
- (iii) $2n^2 + 4n + 1 = 337 \Rightarrow n^2 + 2n - 168 = 0 \Rightarrow (n - 12)(n + 14) = 0$: $\boxed{12}$ (e)
 $\boxed{L = A + D - 1}$

21

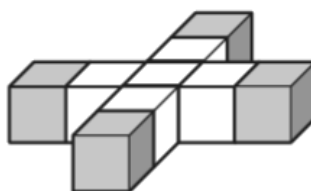
Layer 1



Layer 2



Layer 3



The diagrams show layers of white and grey cubes.
Khadega places these layers on top of each other to make a tower.

(a) Complete the table for towers with 5 and 6 layers.

[4]

Number of layers	1	2	3	4	5	6
Total number of white cubes	0	1	6	15		
Total number of grey cubes	1	5	9	13		
Total number of cubes	1	6	15	28		

(b) (i) Find, in terms of n , the **total** number of **grey** cubes in a tower with n layers.

[2]

(ii) Find the total number of grey cubes in a tower with 60 layers.

[1]

(iii) Khadega has plenty of white cubes but only 200 grey cubes.
How many layers are there in the highest tower that she can build?

[2]

(c) The expression for the **total** number of **white** cubes in a tower with n layers is $pn^2 + qn + 3$.

Question continues on the next page →

Find the value of p and the value of q .
Show all your working.

[5]

- (d) Find an expression, in terms of n , for the **total** number of cubes in a tower with n layers.
Give your answer in its simplest form.

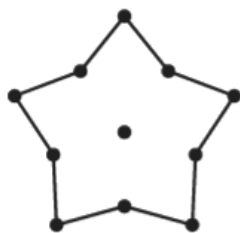
[2]

21 (a) 5 layers: white 28, grey 17, total 45; 6 layers: 45, 21, 66

(b)(i) $4n - 3$ (ii) $4(60) - 3 = 237$ (iii) $4n - 3 \leq 200 \Rightarrow n \leq 50.75$: 50

(c) $n = 1$: $p + q + 3 = 0$; $n = 2$: $4p + 2q + 3 = 1 \Rightarrow p = 2, q = -5$ (d) $2n^2 - 5n + 3 + 4n - 3 = 2n^2 - n$

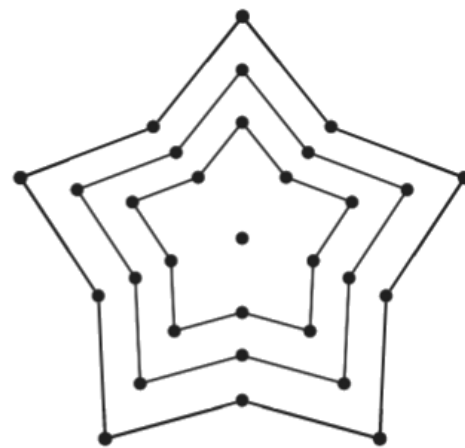
22



Star 1



Star 2



Star 3

The diagrams show a sequence of stars made of lines and dots.

(a) Complete the table for Star 5, Star 7 and Star n .

	Star 1	Star 2	Star 3	Star 4	Star 5		Star 7		Star n
Number of lines	10	20	30	40					
Number of dots	11	21	31	41					

[4]

(b) The sums of the number of dots in two consecutive stars are shown in the table.

Star 1 and Star 2	Star 2 and Star 3	Star 3 and Star 4
32	52	72

Find the sum of the number of dots in

(i) Star 10 and Star 11,

[1]

(ii) Star n and Star $(n + 1)$,

[1]

(iii) Star $(n + 7)$ and Star $(n + 8)$.

[1]

Question continues on the next page →

(c) The **total number of dots** in the first n stars is given by the expression $5n^2 + 6n$.

(i) Show that this expression is correct when $n = 3$.

[2]

(ii) Find the total number of dots in the first 10 stars.

[1]

(d) The total number of dots in the first n stars is $5n^2 + 6n$.

The number of dots in the $(n + 1)$ th star is $10(n + 1) + 1$.

Add these two expressions to show that the total number of dots in the first $(n + 1)$ stars is

$$5(n + 1)^2 + 6(n + 1).$$

You must show each step of your working.

[4]

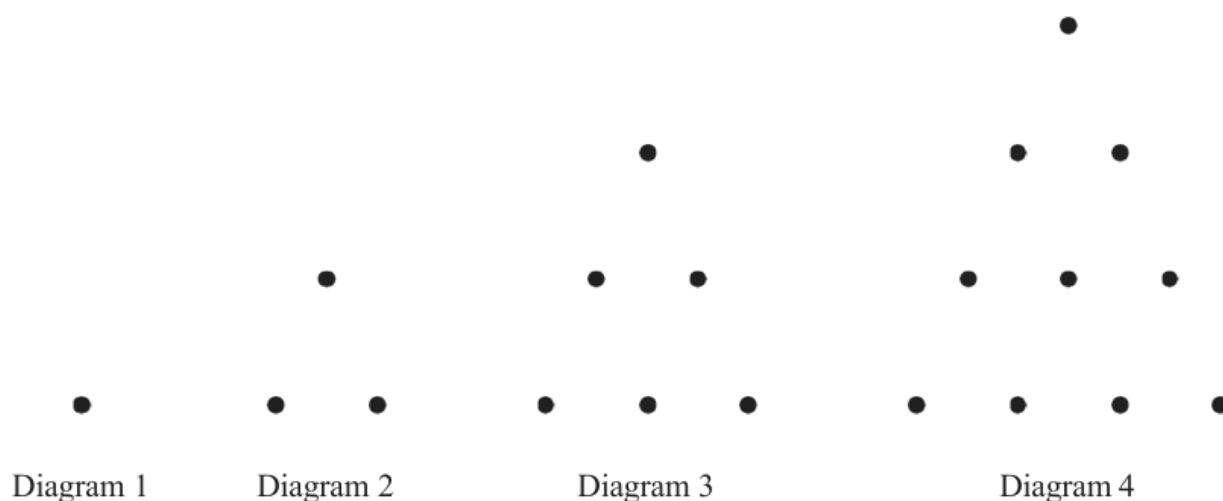
22 (a) Lines: 50, 70, $\boxed{10n}$; Dots: 51, 71, $\boxed{10n + 1}$

(b)(i) $101 + 111 = \boxed{212}$ (ii) $\boxed{20n + 12}$ (iii) $20(n + 7) + 12 = \boxed{20n + 152}$

(c)(i) $5(9) + 6(3) = 63 = 11 + 21 + 31 \checkmark$ (ii) $500 + 60 = \boxed{560}$

(d) $5n^2 + 6n + 10n + 11 = 5n^2 + 16n + 11$; $5(n + 1)^2 + 6(n + 1) = 5n^2 + 10n + 5 + 6n + 6 = 5n^2 + 16n + 11 \checkmark$

23



The first four terms in a sequence are 1, 3, 6 and 10.
They are shown by the number of dots in the four diagrams above.

(a) Write down the next four terms in the sequence. [2]

(b) (i) The sum of the two consecutive terms 3 and 6 is 9. The
sum of the two consecutive terms 6 and 10 is 16.

Complete the following statements using different pairs of terms.

The sum of the two consecutive terms and is

The sum of the two consecutive terms and is [1]

(ii) What special name is given to these sums? [1]

(c) (i) The formula for the n th term in the sequence 1, 3, 6, 10... is $\frac{n(n+1)}{k}$,
where k is an integer.

Find the value of k . [1]

Question continues on the next page →

(ii) Test your formula when $n = 4$, **showing your working.** [1]

(iii) Find the value of the 180th term in the sequence. [1]

(d) (i) Show clearly that the sum of the n th and the $(n + 1)$ th terms is $(n + 1)^2$. [3]

(ii) Find the values of the two consecutive terms which have a sum of 3481. [2]

- 23** (a) $\boxed{15, 21, 28, 36}$ (b)(i) e.g. $10 + 15 = 25$, $15 + 21 = 36$ (ii) $\boxed{\text{square numbers}}$
(c)(i) $\boxed{k = 2}$ (ii) $\frac{4 \times 5}{2} = 10 \checkmark$ (iii) $\frac{180 \times 181}{2} = \boxed{16290}$
(d)(i) $\frac{n(n+1)}{2} + \frac{(n+1)(n+2)}{2} = \frac{(n+1)(2n+2)}{2} = (n+1)^2 \checkmark$ (ii) $(n+1)^2 = 3481 \Rightarrow n = 58$:
 $\boxed{1711 \text{ and } 1770}$

24



Diagram 1

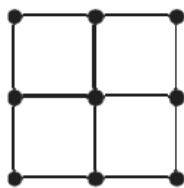


Diagram 2

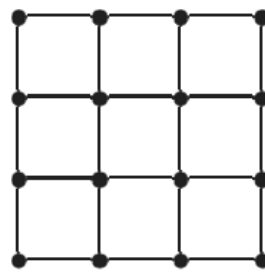


Diagram 3

The first three diagrams in a sequence are shown above.

The diagrams are made up of dots and lines. Each line is one centimetre long.

(a) Make a sketch of the next diagram in the sequence.

[1]

(b) The table below shows some information about the diagrams.

Diagram	1	2	3	4	-----	n
Area	1	4	9	16	-----	x
Number of dots	4	9	16	p	-----	y
Number of one centimetre lines	4	12	24	q	-----	z

(i) Write down the values of p and q .

[2]

(ii) Write down each of x , y and z in terms of n .

[4]

(c) The **total** number of one centimetre lines in the first n diagrams is given by the expression

$$\frac{2}{3}n^3 + fn^2 + gn.$$

(i) Use $n = 1$ in this expression to show that $f + g = \frac{10}{3}$.

[1]

(ii) Use $n = 2$ in this expression to show that $4f + 2g = \frac{32}{3}$.

[2]

(iii) Find the values of f and g .

[3]

(iv) Find the total number of one centimetre lines in the first 10 diagrams.

[1]

- 24** (a) a 4×4 grid of squares. (b)(i) $p = 25, q = 40$ (ii) $x = n^2, y = (n + 1)^2, z = 2n(n + 1)$
- (c)(i) $\frac{2}{3} + f + g = 4 \Rightarrow f + g = \frac{10}{3} \checkmark$ (ii) $\frac{16}{3} + 4f + 2g = 16 \Rightarrow 4f + 2g = \frac{32}{3} \checkmark$
- (iii) $f = 2, g = \frac{4}{3}$ (iv) $\frac{2}{3}(1000) + 200 + \frac{40}{3} = 880$

25 Calculate the next three terms in each of the following sequences:

(a) 1, 3, 4, 7, 11, 18, ...

(b) 4, 9, 13, 22, 35, ...

(c) $\frac{1}{2}$, $\frac{2}{5}$, $\frac{3}{8}$, $\frac{4}{11}$, $\frac{5}{14}$, ...

(d) 5, 15, 45, 135, 405, ...

26 Calculate the first 6 terms of each of the following sequences:

(a) $u_1 = 0$, $u_2 = 3$, $u_n = u_{n-1} + u_{n-2}$

(b) $u_1 = 3$, $u_2 = 4$, $u_n = 2u_{n-1} + u_{n-2}$

(c) $u_1 = 6$, $u_2 = 10$, $u_n = 3u_{n-1} - u_{n-2}$

(d) $u_1 = 1$, $u_2 = 2$, $u_n = u_{n-1} \times u_{n-2}$

25 (a) add the previous two terms: $29, 47, 76$ (b) $57, 92, 149$

(c) $\frac{6}{17}, \frac{7}{20}, \frac{8}{23}$ (d) $\times 3$: $1215, 3645, 10935$

26 (a) $0, 3, 3, 6, 9, 15$ (b) $3, 4, 11, 26, 63, 152$

(c) $6, 10, 24, 62, 162, 424$ (d) $1, 2, 2, 4, 8, 32$

- 27** Calculate the next 3 terms of the sequence,
1, 4, 16, 64, 256, ...
- (b) Determine a formula for the n th term of the sequence.
-
-
-
-
-
-
-
-
-
-
- 28** Determine the formula for the general term of each of the following sequences:
- (a) 15, 75, 375, 1875, 9375, ...
- (b) 1, 3, 9, 27, 81, ...
- (c) 20, 200, 2000, 20 000, 200 000, ...
- (d) 4, 28, 196, 1372, 9604, ...

27 (a) $\boxed{1024, 4096, 16384}$ (multiply by 4) (b) $\boxed{4^{n-1}}$

28 (a) $\boxed{15 \times 5^{n-1}}$ ($= 3 \times 5^n$) (b) $\boxed{3^{n-1}}$ (c) $\boxed{2 \times 10^n}$ (d) $\boxed{4 \times 7^{n-1}}$

- 29** Determine the general formula for the terms of the sequence,
- 1, 7, 13, 19, 25, 31, ...
- (b) Determine the general formula for the terms of the sequence,
- 2, 10, 18, 26, 34, 42, ...
- (c) Determine the general formula for the terms of the sequence,
- $\frac{1}{2}, \frac{7}{10}, \frac{13}{18}, \frac{19}{26}, \frac{25}{34}, \frac{31}{42}, \dots$

- 30** Determine the general formula for the terms of each of the following sequences:

- (a) $\frac{1}{4}, \frac{2}{5}, \frac{3}{6}, \frac{4}{7}, \frac{5}{8}, \dots$
- (b) $\frac{1}{3}, \frac{3}{11}, \frac{5}{19}, \frac{7}{27}, \frac{9}{35}, \dots$
- (c) $\frac{5}{7}, \frac{14}{12}, \frac{23}{17}, \frac{32}{22}, \frac{41}{27}, \dots$
- (d) $\frac{1}{10}, \frac{7}{20}, \frac{13}{40}, \frac{19}{80}, \frac{25}{160}, \dots$

29 (a) $\boxed{6n-5}$ (b) $\boxed{8n-6}$ (c) numerators $6n-5$, denominators $8n-6$: $\boxed{\frac{6n-5}{8n-6}}$

30 (a) $\boxed{\frac{n}{n+3}}$ (b) $\boxed{\frac{2n-1}{8n-5}}$ (c) $\boxed{\frac{9n-4}{5n+2}}$ (d) $\boxed{\frac{6n-5}{5 \times 2^n}}$

31

The table shows the first four terms in sequences A , B , C and D .

Complete the table.

Sequence	1st term	2nd term	3rd term	4th term	5th term		n th term
A	16	25	36	49			
B	5	8	11	14			
C	11	17	25	35			
D	$\frac{3}{2}$	$\frac{4}{3}$	$\frac{5}{4}$	$\frac{6}{5}$			

[12]

31 A: $\boxed{64}$, $\boxed{(n+3)^2}$ B: $\boxed{17}$, $\boxed{3n+2}$

C = A - B: $\boxed{47}$, $\boxed{(n+3)^2 - (3n+2) = n^2 + 3n + 7}$ D: $\boxed{\frac{7}{6}}$, $\boxed{\frac{n+2}{n+1}}$

32

The first four terms of sequences A, B, C and D are shown in the table.

Sequence	1st term	2nd term	3rd term	4th term	5th term		n th term
A	$\frac{1}{3}$	$\frac{2}{4}$	$\frac{3}{5}$	$\frac{4}{6}$			
B	3	4	5	6			
C	-1	0	1	2			
D	-3	0	5	12			

(a) Complete the table.

[8]

(b) Which term in sequence A is equal to $\frac{36}{37}$?

Answer(b) [2]

(c) Which term in sequence D is equal to 725?

Answer(c) [2]

32 (a) A: $\frac{5}{7}$, $\boxed{\frac{n}{n+2}}$; B: 7, $\boxed{n+2}$; C: 3, $\boxed{n-2}$; D = B $\times$ C: 21, $\boxed{n^2-4}$

(b) $\frac{n}{n+2} = \frac{36}{37}$: $\boxed{72}$ (c) $n^2 - 4 = 725 \Rightarrow n^2 = 729$: $\boxed{27}$

33

The table shows the first four terms in sequences A , B , and C .

Sequence	1st term	2nd term	3rd term	4th term	5th term		n th term
A	4	9	14	19			
B	3	10	29	66			
C	1	4	16	64			

Complete the table.

[9]

33 A: $\boxed{24}$, $\boxed{5n-1}$ B: $\boxed{127}$, $\boxed{n^3+2}$ C: $\boxed{256}$, $\boxed{4^{n-1}}$

34 Complete the table for each sequence.

Sequence	1st term	2nd term	3rd term	4th term	5th term	6th term		n th term
A	15	8	1	−6				
B	$\frac{5}{18}$	$\frac{6}{19}$	$\frac{7}{20}$	$\frac{8}{21}$				
C	2	5	10	17				
D	2	6	18	54				

[11]

34 A: $-13, -20, \boxed{22 - 7n}$ B: $\frac{9}{22}, \frac{10}{23}, \boxed{\frac{n+4}{n+17}}$ C: $26, 37, \boxed{n^2 + 1}$ D: $162, 486, \boxed{2 \times 3^{n-1}}$

35

Complete the table for the following sequences.
The first row has been completed for you.

	Sequence				Next two terms	n th term	
	1	5	9	13	17 21	$4n - 3$	
(a)	12	21	30	39			[3]
(b)	80	74	68	62			[3]
(c)	1	8	27	64			[2]
(d)	2	10	30	68			[2]

35 (a) 48, 57; $9n + 3$ (b) 56, 50; $86 - 6n$ (c) 125, 216; n^3 (d) 130, 222;
 $n^3 + n$

36

The table shows four sequences A , B , C and D .

Sequence	1st term	2nd term	3rd term	4th term	5th term		n th term
A	1	8	27	64			
B	5	11	17	23			
C	0.25	0.5	1	2	4		
D	4.75	10.5	16	21			

Complete the table.

[9]

36 A: $\boxed{125}$, $\boxed{n^3}$ B: $\boxed{29}$, $\boxed{6n-1}$ C: $\boxed{2^{n-3}}$ D = B - C: $\boxed{25}$, $\boxed{6n-1-2^{n-3}}$

37

Sequence	1st term	2nd term	3rd term	4th term	5th term		n th term
A	13	9	5	1			
B	0	7	26	63			
C	$\frac{7}{8}$	$\frac{8}{16}$	$\frac{9}{32}$	$\frac{10}{64}$			

(a) Complete the table for the three sequences.

[10]

(b) One term in Sequence C is $\frac{p}{q}$.

Write down the next term in Sequence C in terms of p and q .

..... [2]

37 (a) A: -3 , $17 - 4n$; B: 124 , $n^3 - 1$; C: $\frac{11}{128}$, $\frac{n+6}{2^{n+2}}$ (b) $\frac{p+1}{2q}$

38 The first and the n th terms of sequences A , B and C are shown in the table below.

For
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Use

(a) Complete the table for each sequence.

	1st term	2nd term	3rd term	4th term	5th term	n th term
Sequence A	1					n^3
Sequence B	4					$4n$
Sequence C	4					$(n + 1)^2$

[5]

(b) Find

(i) the 8th term of sequence A ,

Answer(b)(i) [1]

(ii) the 12th term of sequence C .

Answer(b)(ii) [1]

(c) (i) Which term in sequence A is equal to 15 625?

Answer(c)(i) [1]

(ii) Which term in sequence C is equal to 10 000?

Answer(c)(ii) [1]

(d) The first four terms of sequences D and E are shown in the table below.

Use the results from **part (a)** to find the 5th and the n th terms of the sequences D and E .

	1st term	2nd term	3rd term	4th term	5th term	n th term
Sequence D	5	16	39	80		
Sequence E	0	1	4	9		

[4]

38 (a) A: 8, 27, 64, 125; B: 8, 12, 16, 20; C: 9, 16, 25, 36

(b)(i) $\boxed{512}$ (ii) $\boxed{169}$ (c)(i) $\boxed{25}$ (ii) $\boxed{99}$

(d) $D = A+B$: $\boxed{145}$, $\boxed{n^3 + 4n}$; $E = C-B$: $\boxed{16}$, $\boxed{(n+1)^2 - 4n = (n-1)^2}$

- 39 (a) (i) Work out the first 3 terms of the sequence whose n th term is $n(n+2)$.

Answer(a)(i) , , [2]

For
Examiner's
Use

- (ii) Which term in this sequence is equal to 168?

Answer(a)(ii) [3]

- (b) Find a formula for the n th term of the following sequences.

- (i) 5 8 11 14 17

Answer(b)(i) [2]

- (ii) 1 2 4 8 16

Answer(b)(ii) [2]

- (c)



Diagram 1



Diagram 2



Diagram 3

A sequence of diagrams is formed by drawing equilateral triangles each of side one centimetre.
Diagram 1 has 3 one centimetre lines.
Diagram 2 has 9 one centimetre lines.

The formula for the **total** number of one centimetre lines needed to draw **all of the first n diagrams** is

$$an^3 + bn^2 + n.$$

Find the values of a and b .

Answer(c) $a =$

$b =$ [6]

39 (a)(i) $\boxed{3, 8, 15}$ (ii) $n^2 + 2n - 168 = 0 \Rightarrow (n - 12)(n + 14) = 0$: $\boxed{12}$

(b)(i) $\boxed{3n + 2}$ (ii) $\boxed{2^{n-1}}$

(c) Totals: $3, 3 + 9 = 12, 12 + 18 = 30$. $a + b + 1 = 3, 8a + 4b + 2 = 12$
 $\Rightarrow \boxed{a = \frac{1}{2}, b = \frac{3}{2}}$

40

- (a) Complete the table for the 6th term and the n th term in each sequence.

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	Sequence	6th term		n th term
A	11, 9, 7, 5, 3			
B	1, 4, 9, 16, 25			
C	2, 6, 12, 20, 30			
D	3, 9, 27, 81, 243			
E	1, 3, 15, 61, 213			

[12]

- (b) Find the value of the 100th term in

- (i) Sequence A ,

Answer(b)(i) [1]

- (ii) Sequence C .

Answer(b)(ii) [1]

- (c) Find the value of n in Sequence D when the n th term is equal to 6561.

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Question continues on the next page →

Answer(c) n = [1]

(d) Find the value of the 10th term in Sequence *E*.

Answer(d) [1]

- 40** (a) A: 1, $13 - 2n$; B: 36, n^2 ; C: 42, $n(n + 1)$; D: 729, 3^n ; E = D - C: 687, $3^n - n(n + 1)$
- (b)(i) -187 (ii) 10100 (c) $3^n = 6561$: 8 (d) $3^{10} - 110 = 58939$

- 41** The table shows the first five terms of sequences A, B and C.

Sequence	1st term	2nd term	3rd term	4th term	5th term	6th term
A	3	4	5	6	7	
B	0	1	4	9	16	
C	-3	-3	-1	3	9	

(a) Complete the table for the 6th term of each sequence. [2]

(b) Write down the n th term of sequence A. [1]

(c) (i) Find the n th term of sequence B. [2]

(ii) Find the value of n when the n th term of sequence B is 8281. [2]

(d) (i) Find the n th term of sequence C in its simplest form. [2]

(ii) Find the 8th term of sequence C. [1]

(e) The n th term of another sequence D is $\left(-\frac{1}{2}\right)^{n-1}$.

Complete the table for the first four terms of sequence D.

Sequence	1st term	2nd term	3rd term	4th term
D				

[3]

- 41** (a) $\boxed{8, 25, 17}$ (b) $\boxed{n+2}$ (c)(i) $\boxed{(n-1)^2}$ (ii) $n-1=91$: $\boxed{92}$
(d)(i) $C = B - A = (n-1)^2 - (n+2) = \boxed{n^2 - 3n - 1}$ (ii) $64 - 24 - 1 = \boxed{39}$ (e)
 $\boxed{1, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}}$

42

The table shows some terms of several sequences.

Term	1	2	3	4		8	
Sequence P	7	5	3	1		p	
Sequence Q	1	8	27	64		q	
Sequence R	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{3}{4}$	$\frac{4}{5}$		r	
Sequence S	4	9	16	25		s	
Sequence T	1	3	9	27		t	
Sequence U	3	6	7	-2		u	

(a) Find the values of p , q , r , s , t and u . [6]

(b) Find the n th term of sequence

(i) P, [1]

(ii) Q, [1]

(iii) R, [1]

(iv) S, [1]

(v) T, [1]

(vi) U. [1]

(c) Which term in sequence **P** is equal to -777 ? [2]

(d) Which term in sequence **T** is equal to 177 147? [2]

42 (a) $p = -7, q = 512, r = \frac{8}{9}, s = 81, t = 2187, u = 81 - 2187 = -2106$

(b) P $\boxed{9 - 2n}$ Q $\boxed{n^3}$ R $\boxed{\frac{n}{n+1}}$ S $\boxed{(n+1)^2}$ T $\boxed{3^{n-1}}$ U = S - T: $\boxed{(n+1)^2 - 3^{n-1}}$

(c) $9 - 2n = -777$: $\boxed{393}$ (d) $3^{n-1} = 177147 = 3^{11}$: $\boxed{12}$

43

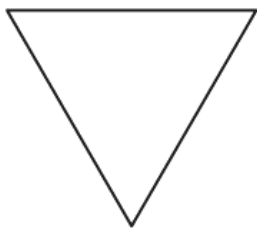


Diagram 1

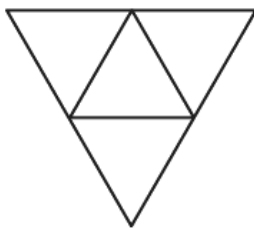


Diagram 2

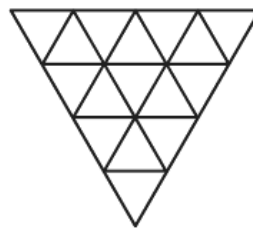


Diagram 3

The first three diagrams in a sequence are shown above.

Diagram 1 shows an equilateral triangle with sides of length 1 unit.

In Diagram 2, there are 4 triangles with sides of length $\frac{1}{2}$ unit.

In Diagram 3, there are 16 triangles with sides of length $\frac{1}{4}$ unit.

(a) Complete this table for Diagrams 4, 5, 6 and n .

[6]

	Diagram 1	Diagram 2	Diagram 3	Diagram 4	Diagram 5	Diagram 6		Diagram n
Length of side	1	$\frac{1}{2}$	$\frac{1}{4}$					
Length of side as a power of 2	2^0	2^{-1}	2^{-2}					

(b) (i) Complete this table for the number of the smallest triangles in Diagrams 4, 5 and 6.

[2]

	Diagram 1	Diagram 2	Diagram 3	Diagram 4	Diagram 5	Diagram 6
Number of smallest triangles	1	4	16			
Number of smallest triangles as a power of 2	2^0	2^2	2^4			

(ii) Find the number of the smallest triangles in Diagram n , giving your answer as a power of 2.

[1]

(c) Calculate the number of the smallest triangles in the diagram where the smallest triangles have sides of

Length $\frac{1}{128}$ unit.

[2]

43 (a) Length: $\frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \boxed{\left(\frac{1}{2}\right)^{n-1}}$; power: $2^{-3}, 2^{-4}, 2^{-5}, \boxed{2^{1-n}}$

(b)(i) $64, 256, 1024 = 2^6, 2^8, 2^{10}$ (ii) $\boxed{2^{2n-2}}$ (c) $\frac{1}{128} = 2^{-7} \Rightarrow n = 8: 2^{14} = \boxed{16384}$

44

- (a) The first five terms P_1, P_2, P_3, P_4 and P_5 of a sequence are given below.

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Use

$$1 = 1 = P_1$$

$$1 + 2 = 3 = P_2$$

$$1 + 2 + 3 = 6 = P_3$$

$$1 + 2 + 3 + 4 = 10 = P_4$$

$$1 + 2 + 3 + 4 + 5 = 15 = P_5$$

- (i) Write down the next term, P_6 , in the sequence 1, 3, 6, 10, 15...

Answer(a)(i) [1]

- (ii) The formula for the n th term of this sequence is

$$P_n = \frac{1}{2}n(n + 1).$$

Show this formula is true when $n = 6$.

Answer (a)(ii)

[1]

- (iii) Use the formula to find P_{50} , the 50th term of this sequence.

Answer(a)(iii) [1]

- (iv) Use your answer to **part (iii)** to find $3 + 6 + 9 + 12 + 15 + \dots + 150$.

Answer(a)(iv) [1]

- (v) Find $1 + 2 + 3 + 4 + 5 + \dots + 150$.

Answer(a)(v) [1]

- (vi) Use your answers to **parts (iv) and (v)** to find the sum of the numbers less than 150 which are **not** multiples of 3.

Answer(a)(vi) [1]

This question continues on the next page.

Question continues on the next page →

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Use

- (b) The first five terms, S_1, S_2, S_3, S_4 and S_5 of a different sequence are given below.

$$\begin{aligned}(1 \times 1) &= 1 = S_1 \\(1 \times 2) + (2 \times 1) &= 4 = S_2 \\(1 \times 3) + (2 \times 2) + (3 \times 1) &= 10 = S_3 \\(1 \times 4) + (2 \times 3) + (3 \times 2) + (4 \times 1) &= 20 = S_4 \\(1 \times 5) + (2 \times 4) + (3 \times 3) + (4 \times 2) + (5 \times 1) &= 35 = S_5\end{aligned}$$

- (i) Work out the next term, S_6 , in the sequence 1, 4, 10, 20, 35...

Answer(b)(i) [2]

- (ii) The formula for the n th term of this sequence is

$$S_n = \frac{1}{6}n(n+1)(n+2).$$

Show this formula is true for $n = 6$.

Answer(b)(ii)

[1]

- (iii) Find $(1 \times 20) + (2 \times 19) + (3 \times 18) + \dots + (20 \times 1)$.

Answer(b)(iii) [1]

- (c) Show that $S_6 - S_5 = P_6$, where P_6 is your answer to **part (a)(i)**.

Answer(c)

[1]

- (d) Show by algebra that $S_n - S_{n-1} = P_n$. $[P_n = \frac{1}{2}n(n+1)]$

Answer(d)

- 44** (a)(i) $\boxed{21}$ (ii) $\frac{1}{2} \times 6 \times 7 = 21$ ✓ (iii) $\boxed{1275}$ (iv) $3 \times 1275 = \boxed{3825}$ (v) $\boxed{11325}$
 (vi) $11325 - 3825 = \boxed{7500}$
 (b)(i) $6 + 10 + 12 + 12 + 10 + 6 = \boxed{56}$ (ii) $\frac{1}{6} \times 6 \times 7 \times 8 = 56$ ✓ (iii) $\frac{1}{6} \times 20 \times 21 \times 22 = \boxed{1540}$
 (c) $56 - 35 = 21 = P_6$ ✓
 (d) $\frac{1}{6}n(n+1)(n+2) - \frac{1}{6}(n-1)n(n+1) = \frac{1}{6}n(n+1)[(n+2) - (n-1)] = \frac{1}{2}n(n+1) = P_n$
 ✓

45

$$1 + 2 + 3 + 4 + 5 + \dots + n = \frac{n(n+1)}{2}$$

(a) (i) Show that this formula is true for the sum of the first 8 natural numbers. [2]

(ii) Find the sum of the first 400 natural numbers. [1]

(b) (i) Show that $2 + 4 + 6 + 8 + \dots + 2n = n(n + 1)$. [1]

(ii) Find the sum of the first 200 even numbers. [1]

(iii) Find the sum of the first 200 odd numbers. [1]

(c) (i) Use the formula at the beginning of the question to find the sum of the first $2n$ natural numbers. [1]

(ii) Find a formula, in its simplest form, for

$$1 + 3 + 5 + 7 + 9 + \dots + (2n - 1).$$

Show your working. [2]

45 (a)(i) $1 + \cdots + 8 = 36$ and $\frac{8 \times 9}{2} = 36$ ✓ (ii) $\boxed{80200}$

(b)(i) $2(1 + 2 + \cdots + n) = 2 \times \frac{n(n+1)}{2} = n(n+1)$ ✓ (ii) $\boxed{40200}$ (iii) $80200 - 40200 = \boxed{40000}$

(c)(i) $\frac{2n(2n+1)}{2} = \boxed{n(2n+1)}$ (ii) $n(2n+1) - n(n+1) = \boxed{n^2}$

46

- (a) (i) The first three positive integers 1, 2 and 3 have a sum of 6.

Write down the sum of the first 4 positive integers.

For
Examiner's
Use

Answer(a)(i) [1]

- (ii) The formula for the sum of the first
- n
- integers is
- $\frac{n(n+1)}{2}$
- .

Show the formula is correct when $n = 3$.

Answer(a)(ii)

[1]

- (iii) Find the sum of the first 120 positive integers.

Answer(a)(iii) [1]

- (iv) Find the sum of the integers

$$121 + 122 + 123 + 124 + \dots + 199 + 200.$$

Answer(a)(iv) [2]

- (v) Find the sum of the even numbers

$$2 + 4 + 6 + \dots + 800.$$

Answer(a)(v) [2]

- (b) (i) Complete the following statements about the sums of cubes and the sums of integers.

$$1^3 = 1$$

$$1 = 1$$

$$1^3 + 2^3 = 9$$

$$1 + 2 = 3$$

For
Examine
Use

Question continues on the next page →

$$1^3 + 2^3 + 3^3 = \dots\dots\dots$$

$$1 + 2 + 3 = \dots\dots\dots$$

$$1^3 + 2^3 + 3^3 + 4^3 = \dots\dots\dots$$

$$1 + 2 + 3 + 4 = \dots\dots\dots [2]$$

- (ii) The sum of the first 14 integers is 105.

Find the sum of the first 14 cubes.

Answer(b)(ii) [1]

- (iii) Use the formula in **part(a)(ii)** to write down a formula for the sum of the first n cubes.

Answer(b)(iii) [1]

- (iv) Find the sum of the first 60 cubes.

Answer(b)(iv) [1]

- (v) Find n when the sum of the first n cubes is 278 784.

Answer(b)(v) $n =$ [2]

46 (a)(i) $\boxed{10}$ (ii) $\frac{3 \times 4}{2} = 6 \checkmark$ (iii) $\boxed{7260}$ (iv) $20100 - 7260 = \boxed{12840}$ (v) $2(1 + \cdots + 400) = \boxed{160400}$

(b)(i) $\boxed{36, 100}$; $\boxed{6, 10}$ (ii) $105^2 = \boxed{11025}$ (iii) $\boxed{\left[\frac{n(n+1)}{2}\right]^2}$ (iv) $1830^2 = \boxed{3348900}$

(v) $\sqrt{278784} = 528 = \frac{n(n+1)}{2} \Rightarrow n^2 + n - 1056 = 0$: $\boxed{32}$

47 (a)

1	= 1
1 + 2	= 3
1 + 2 + 3	= 6
1 + 2 + 3 + 4	= 10

(i) Write down the next line of this pattern. [1]

(ii) The sum of the first n integers is $\frac{n}{k}(n+1)$.

Show that $k=2$.

[2]

(iii) Find the sum of the first 60 integers. [1]

(iv) Find n when the sum of the first n integers is 465. [2]

(v) $1 + 2 + 3 + 4 + \dots + x = \frac{(n-8)(n-7)}{2}$

Write x in terms of n .

[1]

(b)

1^3	= 1
$1^3 + 2^3$	= 9

Question continues on the next page →

$$1^3 + 2^3 + 3^3 = 36$$

$$1^3 + 2^3 + 3^3 + 4^3 = 100$$

(i) Complete the statement.

$$1^3 + 2^3 + 3^3 + 4^3 + 5^3 = \dots\dots\dots = (\dots\dots\dots)^2 \quad [2]$$

(ii) The sum of the first n integers is $\frac{n}{2}(n+1)$.

Find an expression, in terms of n , for the sum of the first n cubes. [1]

(iii) Find the sum of the first 19 cubes. [2]

47 (a)(i) $1 + 2 + 3 + 4 + 5 = 15$ (ii) $n = 1: \frac{1}{k} \times 2 = 1 \Rightarrow k = 2 \checkmark$ (iii) $\boxed{1830}$
(iv) $n^2 + n - 930 = 0 \Rightarrow (n - 30)(n + 31) = 0: \boxed{30}$ (v) $\boxed{x = n - 8}$
(b)(i) $\boxed{225 = (15)^2}$ (ii) $\boxed{\left[\frac{n(n+1)}{2}\right]^2}$ (iii) $190^2 = \boxed{36100}$

48

Total

Row 1	1	=	1
Row 2	3 + 5	=	8
Row 3	7 + 9 + 11	=	27
Row 4	13 + 15 + 17 + 19	=	64
Row 5			
Row 6			

The rows above show sets of consecutive odd numbers and their totals.

(a) Complete Row 5 and Row 6. [2]

(b) What is the special name given to the numbers 1, 8, 27, 64...? [1]

(c) Write down in terms of n ,

(i) how many consecutive odd numbers there are in Row n , [1]

(ii) the total of these numbers. [1]

(d) The first number in Row n is given by $n^2 - n + 1$.

Show that this formula is true for Row 4. [1]

Question continues on the next page →

(e) The total of Row 3 is 27. This can be calculated by $(3 \times 7) + 2 + 4$.

The total of Row 4 is 64. This can be calculated by $(4 \times 13) + 2 + 4 + 6$. The total of Row 7 is 343. Show how this can be calculated in the same way.

[1]

(f) The total of the first n **even** numbers is $n(n + 1)$.

Write down a formula for the total of the first $(n - 1)$ **even** numbers.

[1]

(g) Use the results of **parts (d), (e) and (f)** to show clearly that the total of the numbers in Row n gives your answer to **part (c)(ii)**.

[2]

- 48** (a) Row 5: $21 + 23 + 25 + 27 + 29 = 125$; Row 6: $31 + 33 + 35 + 37 + 39 + 41 = 216$ (b) cube numbers
- (c)(i) n (ii) n^3 (d) $16 - 4 + 1 = 13$ ✓ (e) first number $49 - 7 + 1 = 43$: $(7 \times 43) + 2 + 4 + 6 + 8 + 10 + 12 = 343$
- (f) $(n-1)n$ (g) $n(n^2 - n + 1) + n(n-1) = n^3 - n^2 + n + n^2 - n = n^3$ ✓

49

The first four terms of a sequence are

$$T_1 = 1^2 \quad T_2 = 1^2 + 2^2 \quad T_3 = 1^2 + 2^2 + 3^2 \quad T_4 = 1^2 + 2^2 + 3^2 + 4^2.$$

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Use

- (a) The n th term is given by $T_n = \frac{1}{6} n(n+1)(2n+1)$.

Work out the value of T_{23} .

$$\text{Answer(a)} \quad T_{23} = \dots\dots\dots [2]$$

- (b) A new sequence is formed as follows.

$$U_1 = T_2 - T_1 \quad U_2 = T_3 - T_2 \quad U_3 = T_4 - T_3 \quad \dots\dots$$

- (i) Find the values of U_1 and U_2 .

$$\text{Answer(b)(i)} \quad U_1 = \dots\dots\dots \text{ and } U_2 = \dots\dots\dots [2]$$

- (ii) Write down a formula for the n th term, U_n .

$$\text{Answer(b)(ii)} \quad U_n = \dots\dots\dots [1]$$

- (c) The first four terms of another sequence are

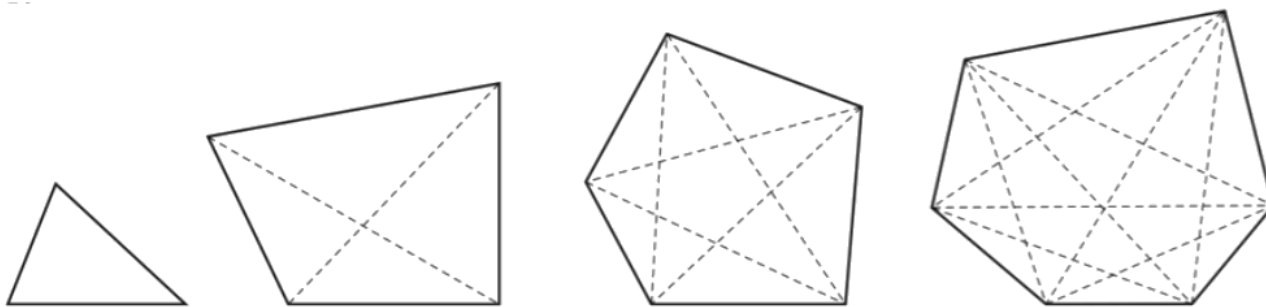
$$V_1 = 2^2 \quad V_2 = 2^2 + 4^2 \quad V_3 = 2^2 + 4^2 + 6^2 \quad V_4 = 2^2 + 4^2 + 6^2 + 8^2.$$

By comparing this sequence with the one in **part (a)**, find a formula for the n th term, V_n .

$$\text{Answer(c)} \quad V_n = \dots\dots\dots [2]$$

49 (a) $\frac{1}{6} \times 23 \times 24 \times 47 = \boxed{4324}$ (b)(i) $U_1 = \boxed{4}$, $U_2 = \boxed{9}$ (ii) $\boxed{(n+1)^2}$
(c) Each term is $2^2 \times$ the matching T term, so $V_n = 4T_n = \boxed{\frac{2}{3}n(n+1)(2n+1)}$

50

Examin
Use

The diagrams show some polygons and their diagonals.

(a) Complete the table.

Number of sides	Name of polygon	Total number of diagonals
3	triangle	0
4	quadrilateral	2
5		5
6	hexagon	9
7	heptagon	14
8		

[3]

(b) Write down the total number of diagonals in

(i) a decagon (a 10-sided polygon),

Answer(b)(i) [1]

(ii) a 12-sided polygon.

Answer(b)(ii) [1]

(c) A polygon with n sides has a total of $\frac{1}{p}n(n - q)$ diagonals, where p and q are integers.

For
Examiner's
Use

(i) Find the values of p and q .

Question continues on the next page →

Answer(c)(i) $p =$

$q =$ [3]

- (ii) Find the total number of diagonals in a polygon with 100 sides.

Answer(c)(ii) [1]

- (iii) Find the number of sides of a polygon which has a total of 170 diagonals.

Answer(c)(iii) [2]

- (d) A polygon with $n + 1$ sides has 30 more diagonals than a polygon with n sides.

Find n .

Answer(d) $n =$ [1]

- 50** (a) 5: pentagon; 8: octagon, $\boxed{20}$ (b)(i) $\boxed{35}$ (ii) $\boxed{54}$
(c)(i) $n = 4$: $\frac{1}{p} \times 4(4 - q) = 2$; $n = 5$: $\frac{1}{p} \times 5(5 - q) = 5 \Rightarrow \boxed{p = 2, q = 3}$ (ii)
 $\frac{1}{2}(100)(97) = \boxed{4850}$
(iii) $n(n - 3) = 340 \Rightarrow (n - 20)(n + 17) = 0$: $\boxed{20}$
(d) $\frac{1}{2}(n + 1)(n - 2) - \frac{1}{2}n(n - 3) = n - 1 = 30$: $\boxed{n = 31}$

Tip: here we do not need the n th-term method — substitute two values of n and solve simultaneously.

51

In all the following sequences, after the first two terms, the rule is to add the previous two terms to find the next term.

*For
Examine
Use*

- (a) Write down the next two terms in this sequence.

1 1 2 3 5 8 13 [1]

- (b) Write down the first two terms of this sequence.

..... 3 11 14 [2]

- (c) (i) Find the value of d and the value of e .

2 d e 10

Answer(c)(i) $d =$

$e =$ [3]

- (ii) Find the value of x , the value of y and the value of z .

-33 x y z 18

Answer(c)(ii) $x =$

$y =$

$z =$ [5]

51 (a) $\boxed{21, 34}$ (b) $\boxed{-5, 8}$ (c)(i) $e = 2 + d, d + e = 10 \Rightarrow \boxed{d = 4, e = 6}$
(ii) $y = x - 33, z = x + y = 2x - 33, y + z = 18 \Rightarrow 3x - 66 = 18: \boxed{x = 28, y = -5, z = 23}$

52

(a) The n th term of a sequence is $n(n+1)$.

(i) Write the two missing terms in the spaces. 2, 6, , 20, [2]

(ii) Write down an expression in terms of n for the $(n + 1)$ th term. [1](iii) The difference between the n th term and the $(n + 1)$ th term is $pn + q$.Find the values of p and q . [2]

(iv) Find the positions of the two consecutive terms which have a difference of 140. [2]

(b) A sequence $u_1, u_2, u_3, u_4, \dots$ is given by the following rules.

$$u_1 = 2, \quad u_2 = 3 \quad \text{and} \quad u_n = 2u_{n-2} + u_{n-1} \text{ for } n \geq 3.$$

For example, the third term is u_3 and $u_3 = 2u_1 + u_2 = 2 \times 2 + 3 = 7$.So, the sequence is 2, 3, 7, u_4 , u_5 ,(i) Show that $u_4 = 13$. [1](ii) Find the value of u_5 . [1]

(iii) Two consecutive terms of the sequence are 3413 and 6827.

Find the term before and the term after these two given terms. [2]

- 52** (a)(i) $\boxed{12, 30}$ (ii) $\boxed{(n+1)(n+2)}$ (iii) $(n+1)(n+2) - n(n+1) = 2n+2$:
 $\boxed{p=2, q=2}$
 (iv) $2n+2 = 140 \Rightarrow n = 69$: $\boxed{69\text{th and } 70\text{th}}$
 (b)(i) $u_4 = 2u_2 + u_3 = 6 + 7 = 13 \checkmark$ (ii) $u_5 = 2(7) + 13 = \boxed{27}$
 (iii) $6827 = 2u + 3413 \Rightarrow \text{before} = \boxed{1707}$; after $= 2(3413) + 6827 = \boxed{13653}$

53 Consecutive integers are set out in rows in a grid.

For
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Use

(a) This grid has 5 columns.

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25
26	27	28	29	30
31	32	33	34	35

a		b
	n	
c		d

The shape drawn encloses five numbers 7, 9, 13, 17 and 19. This is the $n = 13$ shape.

In this shape, $a = 7$, $b = 9$, $c = 17$ and $d = 19$.

(i) Calculate $bc - ad$ for the $n = 13$ shape.

Answer(a)(i) [1]

(ii) For the 5 column grid, $a = n - 6$.

Write down b , c and d in terms of n for this grid.

Answer(a)(ii) $b =$

$c =$

$d =$ [2]

(iii) Write down $bc - ad$ in terms of n .
Show clearly that it simplifies to 20.

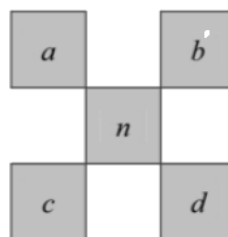
Answer(a)(iii)

Question continues on the next page →

[2]

- (b) This grid has 6 columns. The shape is drawn for $n = 10$.

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30
31	32	33	34	35	36



For
Examiner's
Use

- (i) Calculate the value of $bc - ad$ for $n = 10$.

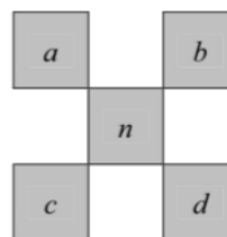
Answer(b)(i) [1]

- (ii) Without simplifying, write down $bc - ad$ in terms of n for this grid.

Answer(b)(ii) [2]

- (c) This grid has 7 columns.

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31	32	33	34	35



Show clearly that $bc - ad = 28$ for $n = 17$.

Answer(c)

[1]

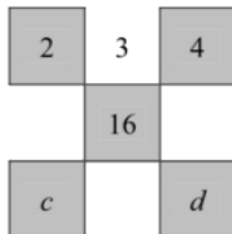
Question continues on the next page →

- (d) Write down the value of $bc - ad$ when there are t columns in the grid.

Answer(d) [1]

For
Examiner
Use

- (e) Find the values of c , d and $bc - ad$ for this shape.



Answer (e) $c =$

$d =$

$bc - ad =$ [2]

- 53** (a)(i) $9 \times 17 - 7 \times 19 = \boxed{20}$ (ii) $b = \boxed{n-4}$, $c = \boxed{n+4}$, $d = \boxed{n+6}$
(iii) $(n-4)(n+4) - (n-6)(n+6) = n^2 - 16 - n^2 + 36 = 20 \checkmark$
(b)(i) $5 \times 15 - 3 \times 17 = \boxed{24}$ (ii) $\boxed{(n-5)(n+5) - (n-7)(n+7)}$
(c) $11 \times 23 - 9 \times 25 = 253 - 225 = 28 \checkmark$ (d) $\boxed{4t}$
(e) $n - a = t + 1 \Rightarrow t = 13$: $c = \boxed{28}$, $d = \boxed{30}$, $bc - ad = 112 - 60 = \boxed{52}$

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1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30
31	32	33	34	35	36

A 3 by 3 square

x	b	c
d	e	f
g	h	i

can be chosen from the 6 by 6 grid above.

(a) One of these squares is

8	9	10
14	15	16
20	21	22

In this square, $x = 8$, $c = 10$, $g = 20$ and $i = 22$.

For this square, calculate the value of

(i) $(i - x) - (g - c)$, [1]

(ii) $cg - xi$. [1]

(b)

x	b	c
d	e	f
g	h	i

(i) $c = x + 2$. Write down g and i in terms of x . [2]

(ii) Use your answers to **part(b)(i)** to show that $(i - x) - (g - c)$ is constant. [1]

(iii) Use your answers to **part(b)(i)** to show that $cg - xi$ is constant. [2]

Question continues on the next page →

(c) The 6 by 6 grid is replaced by a 5 by 5 grid as shown.

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25

A 3 by 3 square

x	b	c
d	e	f
g	h	i

can be chosen from the 5 by 5 grid.

For any 3 by 3 square chosen from this 5 by 5 grid, calculate the **value** of

(i) $(i - x) - (g - c)$, [1]

(ii) $cg - xi$. [1]

(d) A 3 by 3 square is chosen from an n by n grid.

(i) Write down the value of $(i - x) - (g - c)$. [1]

(ii) Find g and i in terms of x and n . [2]

(iii) Find $cg - xi$ in its simplest form. [1]

54 (a)(i) $(22 - 8) - (20 - 10) = \boxed{4}$ (ii) $10 \times 20 - 8 \times 22 = \boxed{24}$

(b)(i) $g = \boxed{x + 12}$, $i = \boxed{x + 14}$ (ii) $14 - (x + 12 - x - 2) = 4 \checkmark$ (iii) $(x + 2)(x + 12) - x(x + 14) = 24 \checkmark$

(c)(i) $\boxed{4}$ (ii) $(x + 2)(x + 10) - x(x + 12) = \boxed{20}$

(d)(i) $\boxed{4}$ (ii) $g = \boxed{x + 2n}$, $i = \boxed{x + 2n + 2}$ (iii) $(x + 2)(x + 2n) - x(x + 2n + 2) = \boxed{4n}$